Rapid Hubble constant inference from GW170817 using GPU-accelerated nested sampling

Ming Yang,[1★] Will Handley,[2,3] and TODO co-authors
[1] St John's College, University of Cambridge, Cambridge CB2 1TP, UK
[2] Institute of Astronomy, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK
[3] Kavli Institute for Cosmology, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK

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ABSTRACT
The binary neutron-star merger GW170817 remains the only confirmed bright gravitational-wave standard siren, and the reference event for direct measurements of the Hubble constant, H_0, from a compact-binary coalescence. The scientific usefulness of a single bright-siren H_0 posterior depends sensitively on the assumed distance, redshift, and host peculiar-velocity priors, and a complete robustness study requires repeated full parameter estimation under each choice. We revisit GW170817 with a GPU-native nested-sampling pipeline that combines a heterodyned likelihood with differentiable JAX waveforms, and exploit its ∼15-minute primary turnaround to carry out a controlled prior-sensitivity, sampler-robustness, and waveform-systematics study. The primary IMRPhenomD_NRTidalv2 baseline run with 5000 live points recovers a broad H_0 posterior with weighted median $79.1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and 68 per cent quantile interval $[68.2, 103.5]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, consistent with the original bright-siren measurement. Replacing the volumetric distance prior with a flat-in-redshift prior by direct sampling shifts the weighted median to $93.6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and raises the tail probability $P(H_0 > 120\,\mathrm{km\,s^{-1}\,Mpc^{-1}})$ from 0.076 to 0.281, while post-hoc reweighting of the baseline samples to the same target prior recovers only 0.195. Targeted mode-isolated runs reveal that the flat-in-redshift posterior is bimodal in (d_L, ι): a compact-distance branch (Mode B, $d_L \in [10, 30]$ Mpc, $H_0^{\rm med} = 130\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) and an extended branch (Mode A, $d_L \in [30, 75]$ Mpc, $H_0^{\rm med} = 78.7\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) carry comparable local evidence after prior-volume normalisation ($\ln \mathcal{B}_{\rm B/A} = -0.66$). Reweighting cannot recover Mode B because the baseline volumetric prior assigns it negligible weight; this is the mechanism behind the reweighting deficit. A four-waveform cross-check (IMRPhenomD_NRTidalv2, IMRPhenomXAS_NRTidalv3, IMRPhenomPv2 without tides, TaylorF2) finds the H_0 median spread within $3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $P(H_0 > 120)$ in $[0.034, 0.137]$, smaller than the prior-induced spread. Sampler-hyperparameter ($n_{\rm delete}/n_{\rm live} \in [0.10, 0.75]$, $n_{\rm bins} \in [251, 1001]$), PSD-source, reference-parameter, and host peculiar-velocity ($\sigma_{v_p} = 150 \rightarrow 250\,\mathrm{km\,s^{-1}}$) variations all shift H_0 by less than $2\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ in median. The primary 5000-live-point analysis completes in ∼14 min on a single NVIDIA A100 GPU, with the heterodyned likelihood providing a 33× ($n_{\rm live} = 500$) to 82× ($n_{\rm live} = 2500$) wall-clock speedup over the unheterodyned reference, making direct prior, waveform, and sampler-robustness checks practical for bright-siren cosmology.

Key words: gravitational waves – methods: data analysis – cosmological parameters – stars: neutron – software: data analysis

1 INTRODUCTION

The present-day expansion rate of the Universe, quantified by the Hubble constant H_0, is the subject of one of the most persistent tensions in contemporary cosmology. Measurements anchored in the early Universe, derived from cosmic-microwave-background observations assuming flat ΛCDM, favour a relatively low value (Planck Collaboration 2016), while late-Universe distance-ladder measurements based on Cepheids and Type Ia supernovae find a significantly higher value (Riess et al. 2016, 2022). The statistical and systematic discrepancy between these two families of measurement is now established at several σ, and its resolution, whether through unrecognised systematics or new cosmological physics, motivates independent probes of the expansion rate.

Gravitational-wave standard sirens are such a probe. The waveform amplitude of a compact-binary merger fixes the luminosity distance directly, without reference to an astrophysical distance ladder (Schutz 1986). When the merger is accompanied by an electromagnetic counterpart that identifies a host galaxy, the host redshift can be combined with the gravitational-wave distance to infer H_0 from a single event. The binary neutron-star merger GW170817 (Abbott et al. 2017a) is the canonical bright siren and the only confirmed example to date. Its association with the kilonova AT2017gfo in NGC 4993 provided the host-galaxy redshift needed to yield the first gravitational-wave measurement of H_0 (Abbott et al. 2017b).

For a bright siren, the H_0 uncertainty is dominated by the luminosity-distance–inclination degeneracy: a face-on binary at larger d_L can produce the same observed strain as a more inclined binary at smaller d_L, with the two solutions mapping to different values of $H_0 = v_{\rm r}/d_L$ for a fixed host recessional velocity $v_{\rm r}$. The

★ E-mail: TODO

resulting posterior is broad, skewed, and only weakly informative on H_0 from one event. In such a regime, the measurement is more sensitive to prior choices than the statistical uncertainty alone would suggest, because the prior shapes the tails of the posterior rather than just its peak. The original GW170817 analysis assumed a volumetric distance prior $\pi(d_L) \propto d_L^2$ and reported that switching to a flat-in-redshift prior, implemented by reweighting posterior samples, produced only a minor change in the inferred H_0 posterior (Abbott et al. 2017b).

Reweighting is attractive because it avoids rerunning expensive Bayesian inference. For a source-parameter posterior $p(\theta \mid d, \pi_0)$ obtained under baseline prior π_0, the target posterior under an alternative prior π_1 can be estimated by assigning each posterior sample a weight proportional to $\pi_1(\theta)/\pi_0(\theta)$. This procedure is statistically consistent provided that the baseline samples cover the support of the target posterior; if π_1/π_0 is large in regions that the baseline run has sampled sparsely, the reweighted estimator has high variance and systematically low tail mass (see e.g. Speagle 2020, for a general discussion of reweighting in the nested-sampling context). For GW170817, the baseline volumetric prior $\pi_0(d_L) \propto d_L^2$ places little mass at the low d_L values corresponding to the top of the H_0 prior range; a flat-in-redshift target prior up-weights exactly this region, and the validity of reweighting is not obvious a priori.

Testing reweighting against a direct posterior obtained by sampling under the target prior requires repeated full gravitational-wave parameter estimation. In CPU-based frameworks such as BILBY (Ashton et al. 2019) and its parallelised counterpart PARALLEL_BILBY, a single 128 s binary-neutron-star analysis with phase-marginalised frequency-domain likelihood costs hundreds of CPU core-hours to thousands of core-hours, depending on waveform, live-point count, and prior bounds. Systematic exploration of distance, velocity, and waveform assumptions therefore becomes prohibitive. This computational cost is not only a practical nuisance: it directly limits the robustness studies that can be performed for a bright-siren cosmological result.

Several recent developments make this constraint less severe. Relative binning, also known as the heterodyned likelihood (Cornish 2013; Zackay et al. 2018), reduces the effective frequency grid for a long inspiral to a small number of bins with negligible loss of accuracy. Differentiable waveform libraries in JAX (Bradbury et al. 2018), including RIPPLE (Edwards et al. 2023), enable end-to-end GPU evaluation of the likelihood. GPU-native nested-sampling kernels (Cabezas et al. 2024; Yallup et al. 2025; Prathaban et al. 2025) retain the evidence estimates that nested sampling provides for free while allowing large live-point counts and repeated full reruns. Related efforts have demonstrated accelerated binary-neutron-star inference with related tools (Krishna et al. 2023; Wong et al. 2023; Wouters et al. 2024), and have underlined the practical importance of compute-efficient parameter estimation for the third-generation detector era (Hu & Veitch 2025).

In this paper we use a GPU-native nested-sampling pipeline, built on BLACKJAX-NS (Yallup et al. 2025; Prathaban et al. 2025) and a JAX implementation of the heterodyned likelihood, to revisit the GW170817 bright-siren H_0 inference. Our focus is not a new single-event measurement of H_0, but a controlled, multi-axis assessment of how that measurement depends on its priors, on the waveform, on the sampler hyperparameters, and on auxiliary inputs (PSDs, heterodyne reference, host peculiar-velocity uncertainty). We first validate the pipeline against GW150914 (using IMRPhenomXPHM, the LVK production waveform) and against reference GW170817 posteriors. We then report the baseline H_0 posterior, compare it with a direct flat-in-redshift posterior and with the reweighted flat-in-redshift posterior, and use targeted mode-isolated runs to identify the (d_L, ι) bimodality that mechanistically explains the reweighting deficit. A four-waveform cross-check (IMRPhenomD_NRTidalv2, IMRPhenomXAS_NRTidalv3, IMRPhenomPv2 without tides, TaylorF2) bounds the residual waveform systematic on H_0. A separate set of robustness sweeps quantifies the sensitivity to nested-sampling hyperparameters, heterodyne-bin count, PSD source, and reference parameters. We finish by reporting runtime and live-point scaling on a single A100 GPU. Performance is presented to motivate that this multi-axis study is now tractable, not as a central claim of the paper.

Section 2 describes the inference setup. Section 3 reports validation. Section 4 contains the H_0 inference, the prior-sensitivity analysis, the bimodality characterisation, the GW-only versus standard-siren comparison, and the multi-waveform cross-check. Section 5 reports runtime, live-point scaling, and the sampler/PSD robustness sweeps. Section 6 discusses implications and limitations. Section 7 concludes.

2 METHOD

2.1 Nested sampling

For data d and source parameters θ under model $\mathcal{M}$, we target the posterior

$$p(\theta \mid d, \mathcal{M}) = \frac{\mathcal{L}(d \mid \theta, \mathcal{M})\, \pi(\theta \mid \mathcal{M})}{Z(d \mid \mathcal{M})}, \tag{1}$$

where $\mathcal{L}$ is the gravitational-wave likelihood, π the prior, and Z the Bayesian evidence. We estimate Z and draw posterior samples by nested sampling (Skilling 2006), using the GPU-native slice sampler implemented in BLACKJAX-NS (Yallup et al. 2025; Prathaban et al. 2025). The sampler evolves a population of live points through batched slice updates; this vectorisation is the key ingredient that keeps the GPU saturated when the per-evaluation likelihood cost is small.

All heterodyned GW170817 runs in this paper use $n_{\rm live} = 5000$, $n_{\rm delete} = 2500$ points removed per iteration ($\frac{1}{2}\, n_{\rm live}$), and $5\, n_{\rm dim}$ MCMC steps per slice; runs terminate when the fractional evidence increment is below 10^{-3} [Source: `paper_knowledge_base/a100_run_data.md`, `paper_knowledge_base/codebase_reference.md`]. Scaling experiments vary $n_{\rm live}$ while holding all other settings fixed.

2.2 Heterodyned likelihood

The GW170817 signal is long enough that the full frequency-domain likelihood uses 259 201 bins over 20–2048 Hz with $\Delta f = 1/128$ Hz for the 128 s strain segment [Source: `paper_knowledge_base/codebase_reference.md`]. Evaluating the likelihood on this grid for every live-point update is prohibitively expensive when millions of evaluations are required. We therefore use a heterodyned (relative-binning) likelihood (Cornish 2013; Zackay et al. 2018), in which the ratio between a candidate waveform and a reference waveform is approximated by a piecewise-linear function over a small set of coarse frequency bins. We use 501 nominal heterodyne bins, of which the effective count is 442 for TaylorF2 and 501 for IMRPhenomD_NRTidalv2 after waveform-specific frequency-support masks are applied [Source: `paper_knowledge_base/a100_run_data.md`]; for GW150914, 383 bins are used [Source: same]. The reference waveform uses the GWTC-1 median posterior parameters for GW170817.

We compare heterodyned and unheterodyned posteriors directly in Section 3 to verify that the approximation does not bias the source-parameter posterior at the level needed for H_0 inference. All reported heterodyned runs also marginalise analytically over the coalescence phase, so the sampled parameter space is fourteen-dimensional for GW170817 (intrinsic masses, spin z-components, tidal deformabilities, inclination, luminosity distance, coalescence time, polarisation angle, sky position, and the cosmological parameters H_0 and v_p), and ten-dimensional for GW150914.

2.3 Joint Hubble-constant likelihood

We sample the Hubble constant jointly with the gravitational-wave source parameters rather than post-processing the d_L marginal. The GW170817 host galaxy NGC 4993 provides a CMB-frame recessional velocity v_r and a Hubble-flow peculiar-velocity estimate $\langle v_p \rangle$; both are used as Gaussian likelihoods on the model prediction. The combined log-likelihood is

$$\ln \mathcal{L}_{\rm tot}(\theta, H_0, v_p) = \ln \mathcal{L}_{\rm GW}(d \mid \theta) + \ln p(v_{\rm r} \mid H_0\, d_L + v_p) + \ln p(\langle v_p \rangle \mid v_p), \quad (2)$$

where d_L is a function of θ only, and H_0 and v_p enter through the velocity terms alone. This is the standard GW170817 bright-siren setup, following Abbott et al. (2017b).

2.4 Priors and prior variants

The baseline prior configuration follows the GW170817 bright-siren analysis: $\pi(d_L) \propto d_L^2$ over $[10, 75]$ Mpc, $\pi(H_0) \propto H_0^{-1}$ over $[45, 250]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and $\pi(v_p)$ uniform over $[-1000, 1000]\,\mathrm{km\,s^{-1}}$ with a Gaussian centred on $\langle v_p \rangle$ at $\sigma_{v_p} = 150\,\mathrm{km\,s^{-1}}$ in the peculiar-velocity likelihood term. Source-parameter priors are the constrained "host-localised" set implemented in `GW170817_heterodyned_1.py`: narrow ranges on chirp mass, mass ratio, and aligned-spin z-components around the GWTC-1 reference values, a uniform tidal prior, and a narrow sky-location prior centred on NGC 4993 [Source: `paper_knowledge_base/codebase_reference.md`].

We consider three prior variants in addition to the baseline:

(i) *Direct flat-in-redshift*: $\pi(z)$ uniform within the corresponding distance range, converted to a d_L prior via a flat ΛCDM cosmology and sampled directly (`GW170817_heterodyned_2.py`).

(ii) *Reweighted flat-in-redshift*: the baseline posterior samples reweighted by the analytic ratio of the flat-in-redshift to volumetric d_L priors (`Plots/reweight_dL_to_flat_z.py`).

(iii) *Enlarged peculiar-velocity uncertainty*: identical to the baseline except $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$ (`GW170817_heterodyned_3.py`).

The direct and reweighted flat-in-redshift posteriors use the same target prior, and differ only in whether the prior is imposed during sampling or applied post-hoc.

2.5 Waveforms

Our primary gravitational-wave model is IMRPhenomD_NRTidalv2, an aligned-spin tidal waveform available in the differentiable RIPPLE library (Edwards et al. 2023). To bound the residual waveform systematic on H_0 we run three additional models: IMRPhenomXAS_NRTidalv3, an upgrade with the most recent NR-calibrated tidal phase prescription on a more recent aligned-spin base; IMRPhenomPv2, a precessing BBH waveform without tides, used to bracket the precession contribution that no precessing tidal waveform in the present RIPPLE inventory provides; and TaylorF2, a point-particle post-Newtonian inspiral-only model used as a waveform-family check. IMRPhenomPv2 runs use a low-spin prior (spin magnitudes uniform in $[0, 0.05]$, isotropic tilts, uniform azimuths) consistent with Abbott et al. (2017b); tidal deformabilities are not sampled.

The original GW170817 bright-siren analysis used IMRPhenomPv2_NRTidal, which includes both precession and NR-calibrated tides. RIPPLE does not currently expose a precessing tidal waveform; we therefore cannot run a like-for-like reproduction of the LVK setup, and instead bracket the relevant systematics with the IMRPhenomXAS_NRTidalv3/IMRPhenomPv2 pair. The agreement of H_0 posteriors across IMRPhenomD_NRTidalv2, IMRPhenomXAS_NRTidalv3, IMRPhenomPv2, and TaylorF2 (Section 4.5) is the quantitative content of the waveform-systematic bound.

For the GW150914 validation we use IMRPhenomXPHM, the LVK GWTC-2.1+ production waveform for that event, which incorporates precession through a multi-scale-analysis treatment and the dominant higher-order modes; this replaces the IMRPhenomD validation used in earlier drafts of this work and turns the validation into a direct LVK reproduction rather than an offset comparison.

Table 1. Heterodyned GW150914 validation with IMRPhenomXPHM and 5000 live points. MAP, weighted median, and weighted 68 per cent quantile intervals are shown.

Parameter	MAP	Median	68% interval
$\mathcal{M}_c/M_\odot$	30.51	30.34	[29.40, 31.37]
q	0.91	0.87	[0.74, 0.96]
d_L/Mpc	479.0	459.8	[365.7, 540.0]
ι/rad	2.64	2.62	[2.34, 2.86]

[Source: `Results/test_suite/gw150914_waveform_comparison.csv`.]

3 VALIDATION

3.1 GW150914

We validate the pipeline on the binary-black-hole event GW150914 (Abbott et al. 2016), which has a short signal and established public posteriors. We analyse the GWOSC strain data with IMRPhenomXPHM and GWTC-2.1 PSDs, using the same heterodyned sampler as for GW170817 with 5000 live points and phase marginalisation. Table 1 summarises the recovered parameters. Chirp mass, luminosity distance, mass ratio, and inclination are all recovered at values consistent with the GWTC-2.1 reference. Figure 1 shows the corresponding corner plot. The sampler returns $\ln Z = 261.09 \pm 0.08$ [Source: `Results/test_suite/gw150914_waveform_comparison.csv`].

The earlier `IMRPhenomD` aligned-spin validation gave $d_L^{\rm med} \approx 405$ Mpc, offset from the LVK reference (~430 Mpc) by approximately one quarter of the 68 per cent posterior width. Switching to IMRPhenomXPHM, which incorporates precession and dominant higher-order modes, produces $d_L^{\rm med} = 459.8$ Mpc with 68 per cent quantile interval $[365.7, 540.0]$ Mpc, in agreement with the GWTC-2.1 reference. The validation is therefore now a direct LVK reproduction.

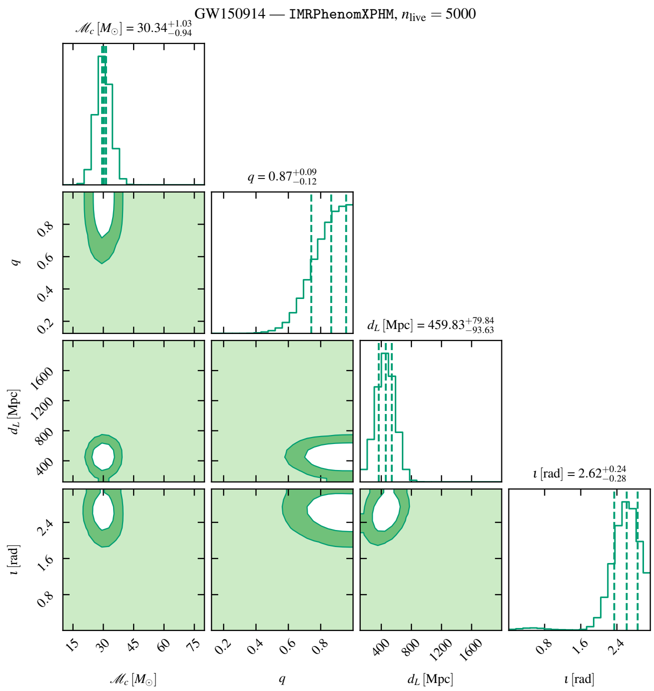

Figure 1. GW150914 validation: heterodyned-pipeline posterior under IMRPhenomXPHM for the principal source parameters. Run `s06__gw150914__imrphenomxphm__lvkbounds__seed0000`; produced by `mnras_paper/figures/plot_all.py:fig_gw150914_corner`.

3.2 Heterodyned versus unheterodyned consistency on GW170817

We next check that the heterodyned approximation does not bias the GW170817 source-parameter posterior. We run the same sampler on the full 259 201-bin frequency-domain likelihood with host-localised priors for both IMRPhenomD_NRTidalv2 and TaylorF2 at $n_{\rm live} = 1500$ to keep the runtime tractable, and additionally at $n_{\rm live} \in \{500, 2500\}$ for IMRPhenomD_NRTidalv2 to verify that the comparison is not live-point limited [Source: `Results/test_suite/unhet_scaling_summary.csv`, `paper_knowledge_base/codebase_reference.md`]. Table 2 compares key marginals. The heterodyned baseline and the $n_{\rm live} = 1500$ unheterodyned run agree on the chirp mass to within the sampler width, and on the weighted median distance to better than one megaparsec. The heterodyned weighted median H_0 is $79.1\,{\rm km\,s^{-1}\,Mpc^{-1}}$, compared with $77.8\,{\rm km\,s^{-1}\,Mpc^{-1}}$ for the unheterodyned run, a shift that is much smaller than the statistical width of the H_0 posterior. The two larger unheterodyned cross-checks ($n_{\rm live} = 500$, $H_0^{\rm med} = 79.5\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $\ln Z = 485.75 \pm 0.30$, 7 317 s; $n_{\rm live} = 2500$, $H_0^{\rm med} = 78.1\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $\ln Z = 490.51 \pm 0.14$, 37 240 s) place the heterodyned median between the two unheterodyned estimates, well within the sampler-statistical width. Figure 2 shows the corresponding corner comparison. We conclude that the heterodyned approximation is adequate for H_0 inference at this event's signal-to-noise ratio; the corresponding wall-clock reduction is quantified in Section 5.2.

3.3 Cross-waveform consistency

Figure 3 compares the IMRPhenomD_NRTidalv2 and TaylorF2 heterodyned posteriors in the host-localised baseline configuration with the GWTC-1 reference samples. The JAX-based analyses recover the same distance–inclination structure as the LVK reference, with the known small offsets expected from the different waveform families and prior bounds. Jensen–Shannon divergences between the IMRPhenomD_NRTidalv2 and TaylorF2 baseline posteriors are 0.0047 nats for H_0 and 0.0057 nats for d_L [Source: `Results/gwtc1_phasemarg/waveform_systematics.csv`], an order of magnitude smaller than the JSD values observed across prior variants in Section 4.2.

To bound the residual waveform systematic on H_0 we add two further models on the LVK-bounds prior set ($M_{\rm comp} \in [0.87, 1.74]\,M_\odot$): IMRPhenomXAS_NRTidalv3, the most recent NR-calibrated tidal model on a modern aligned-spin base, and IMRPhenomPv2, a precessing BBH waveform without tides. Table 3 reports H_0 summaries and evidences for the four-waveform suite. The weighted medians cluster within $3\,{\rm km\,s^{-1}\,Mpc^{-1}}$ across all four waveforms, and the high-H_0 tail probabilities span $[0.034, 0.137]$, with IMRPhenomXAS_NRTidalv3 producing the tightest tail and

Table 2. Heterodyned versus unheterodyned GW170817 posteriors with IMRPhenomD_NRTidalv2, host-localised priors. Weighted medians and 68 per cent quantile intervals are shown.

Parameter	Heterodyned	Unheterodyned
H_0 median / ${\rm km\,s^{-1}\,Mpc^{-1}}$	79.1	77.8
H_0 68% / ${\rm km\,s^{-1}\,Mpc^{-1}}$	[68.2, 103.5]	[67.3, 100.9]
d_L median /Mpc	38.1	38.9
d_L 68% /Mpc	[29.2, 43.7]	[30.0, 44.3]
ι median /rad	2.55	2.55

[Source: `Results/gwtc1_phasemarg/summary_stats.csv`; heterodyned run is the IMRPhenomD_NRTidalv2 baseline and unheterodyned run is `PhaseMarg_Unheterodyned_IMRPhenomD_NRTidalv2_local_psd-gwtc1.csv`.]

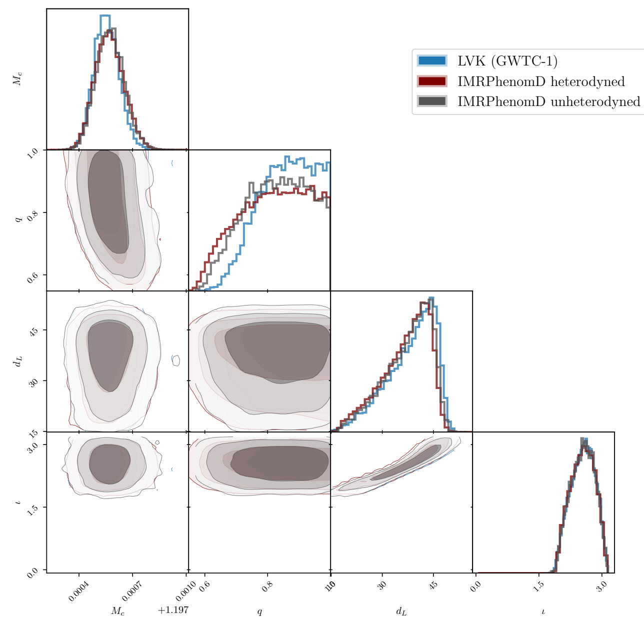

Figure 2. GW170817 posterior comparison between the heterodyned baseline (501 bins, 5000 live points) and an unheterodyned reference run (full 259 201 bins, 1500 live points) under identical host-localised priors. GWTC-1 reference samples are overlaid for context. Produced by `Plots/plot_corner_IMRPhenomD_hetero_vs_unhetero.py`.

Table 3. GW170817 H_0 summaries and nested-sampling evidences for the four-waveform cross-check on the LVK-bounds baseline prior, $n_{\rm live} = 5000$. H_0 values are in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Waveform	MAP	Median	68% q.i.	$P(H_0 > 120)$	$\ln Z$
IMRPhenomD_NRTidalv2	73.0	79.0	[68.4, 102.5]	0.070	489.06 ± 0.10
IMRPhenomXAS_NRTidalv3	71.2	78.2	[68.1, 98.1]	0.034	490.08 ± 0.10
IMRPhenomPv2 (no tides)	71.2	80.7	[68.8, 114.4]	0.137	491.00 ± 0.09
TaylorF2 †	70.1	77.7	[66.8, 101.9]	0.065	486.90 ± 0.10

[Source: `Results/test_suite/gw170817_waveform_comparison.csv`, `Results/gwtc1_phasemarg/summary_stats.csv`, `Results/gwtc1_phasemarg/evidence_table.csv`.]
† TaylorF2 is on the host-localised prior set; the other three are on LVK-bounds. The $\ln Z$ values are not directly comparable across these prior sets, but the H_0 summaries are.

IMRPhenomPv2 the broadest. The full H_0 kernel-density estimates (Figure 4) overlay closely between 40 and $\sim 110\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and disagree mainly in the tail. The waveform-induced shift in $P(H_0 > 120)$ (~ 0.10 across the four waveforms) remains smaller than the prior-induced shift documented in Section 4.2 ($0.076 \to 0.281$), and *a fortiori* smaller than the difference between direct and reweighted flat-in-redshift estimates (0.281 versus 0.195).

4 RESULTS

4.1 Baseline Hubble-constant posterior

Under the baseline volumetric distance prior, the IMRPhenomD_NRTidalv2 analysis yields a broad, asymmetric H_0 posterior. The maximum-posterior value is

$$H_0^{\rm MAP} = 71.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{3}$$

with weighted 68 per cent quantile interval

$$H_0 \in [68.2, 103.5]\,\mathrm{km\,s^{-1}\,Mpc^{-1}} \quad (68\%\ \text{q.i.}), \tag{4}$$

and weighted median $79.1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [Source: `Results/gwtc1_phasemarg/summary_stats.csv`]. A kernel-density estimate of the 68 per cent highest-posterior-density interval gives $[62.7, 91.1]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; we report both the weighted-quantile and KDE-HPD forms because the posterior is strongly asymmetric and the two statistics emphasise different parts of the distribution. Figure 5 shows the baseline posterior with the LVK (Abbott et al. 2017b) 68 per cent reference band for context.

This posterior reproduces the qualitative features of the original GW170817 bright-siren measurement (Abbott et al. 2017b): the peak is in the neighbourhood of the distance-ladder value, the posterior is broad enough to remain consistent with CMB measurements, and the high-H_0 tail extends to the upper edge of the prior range. The present analysis does not sharpen the single-event H_0 constraint relative to the literature; the purpose of reporting this posterior is to establish the baseline against which the prior-sensitivity comparison in Section 4.2, the bimodality characterisation in Section 4.3, and the multi-waveform comparison summarised in Figure 6 are performed.

4.2 Prior sensitivity and the reweighting test

The central scientific result of this paper is a direct comparison between the flat-in-redshift H_0 posterior obtained by sampling and that obtained by reweighting the baseline posterior to the same target prior. Table 4 summarises the four IMRPhenomD_NRTidalv2 H_0 posteriors.

The direct flat-in-redshift run shifts the weighted median from 79.1 to $93.6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and broadens the 68 per cent quantile interval by about a factor of 2 on the high side. The reweighted flat-in-redshift posterior moves in the same direction but by a smaller amount: its weighted median is $88.9\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and its 68 per cent quantile interval reaches $126.2\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ rather than $146.9\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The difference is sharpest in the high-H_0 tail: the probability

$$P(H_0 > 120\,\mathrm{km\,s^{-1}\,Mpc^{-1}}) = 0.076,\ 0.195,\ 0.281$$

for the baseline, reweighted and directly sampled posteriors respectively; $P(H_0 > 150\,\mathrm{km\,s^{-1}\,Mpc^{-1}})$ rises from 0.016 in the baseline to 0.056 in the reweighted posterior and 0.147 in the direct posterior [Source: `Results/gwtc1_phasemarg/prior_sensitivity.csv`]. The earth-mover (Wasserstein-1) distance of each posterior from the baseline is $11.2\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for reweighting and $20.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for direct sampling. In the sense of cumulative distribution, the reweighted posterior is therefore only about half of the way to the direct flat-in-redshift posterior. Figure 7 visualises this hierarchy.

The mechanism is directly visible in the luminosity-distance marginal, shown in Figure 8. A flat-in-redshift prior up-weights small d_L relative to the baseline volumetric prior; small d_L in turn maps to large H_0 for the fixed host recessional velocity. In the direct run, the sampler explores the low-d_L region with live points drawn from the flat-in-redshift prior, and the posterior accumulates weight there. The probability $P(d_L < 30\,\mathrm{Mpc})$ rises from 0.179 in the baseline to 0.428 in the direct flat-in-redshift posterior. In the reweighted run, the baseline samples are re-scored by the prior ratio, but there are few baseline samples at low d_L because the baseline volumetric prior had placed little probability there in the first place. The reweighted estimator consequently under-represents the low-d_L region ($P(d_L < 30\,\mathrm{Mpc}) = 0.357$) and, as a consequence, under-represents the high-H_0 region. Corner comparisons in the d_L–ι plane confirm that the direct and reweighted posteriors differ mainly in the low-d_L, small-ι corner; the waveform-parameter marginals are almost identical [Source: `Results/gwtc1_phasemarg/plots/corner_reweighted_vs_sampled_fla` produced by `Plots/plot_corner_reweighted_vs_sampled_flatZ.py`].

The enlarged peculiar-velocity variant ($\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$) shifts the H_0 posterior only slightly: the median falls marginally to $78.6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and the high-H_0 tail thins, giving $P(H_0 > 120) = 0.067$. The Wasserstein distance from the baseline is $1.24\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, an order of magnitude smaller than either flat-in-redshift comparison. For GW170817 with NGC 4993 identified, the dominant prior-induced uncertainty in H_0 is therefore the distance prior, not the peculiar-velocity uncertainty.

We caution against over-interpreting these shifts as physical results for H_0. None of them changes the qualitative statement that GW170817 alone is consistent with both early- and late-Universe measurements of the expansion rate. The scientific content is that the magnitude of the shift under a change of distance prior is a property of the inference, not of the cosmology, and that the magnitude one would report using reweighting alone is materially different from the magnitude recovered by direct sampling.

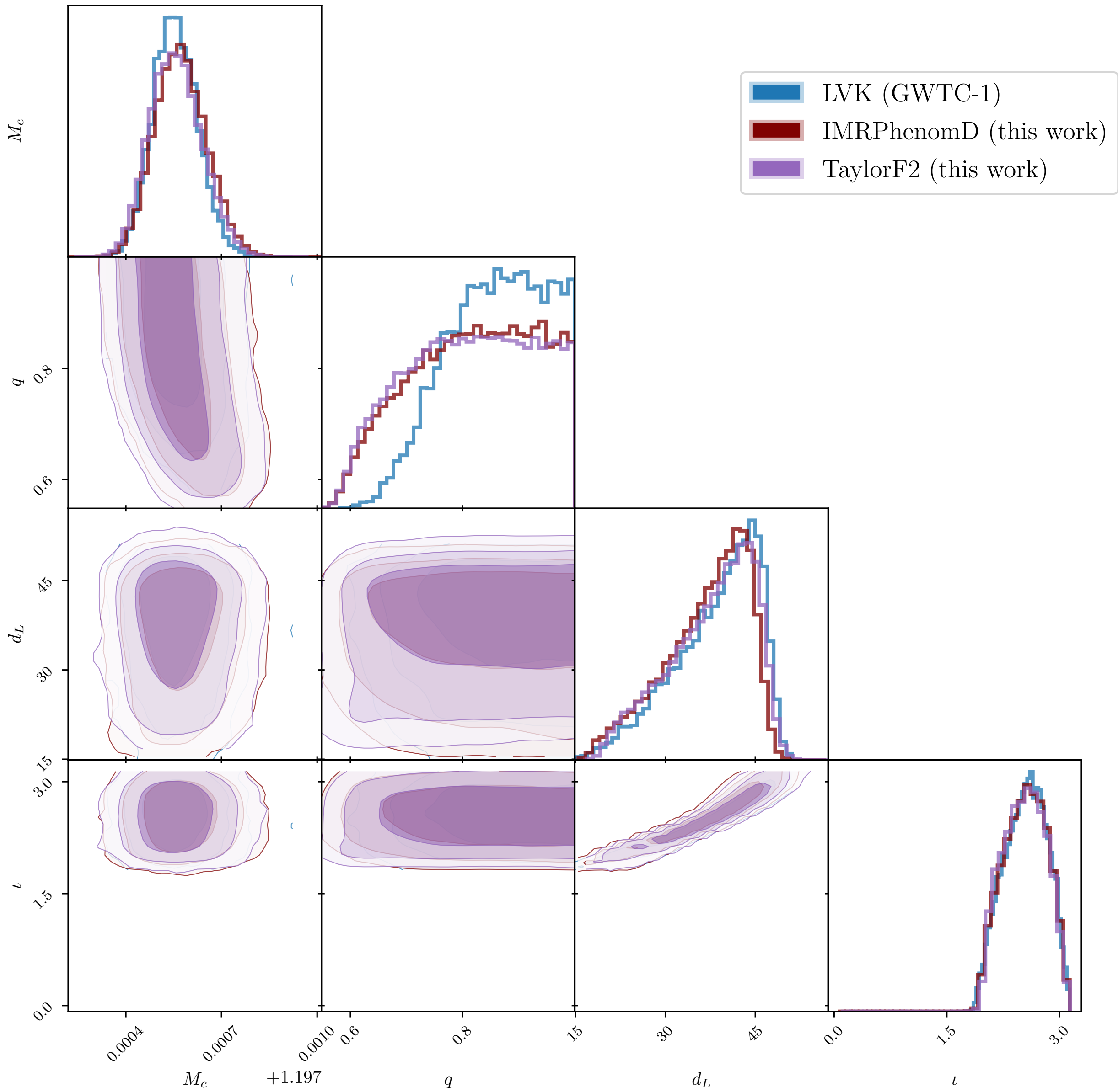

Figure 3. GW170817 source-parameter posteriors under IMRPhenomD_NRTidalv2 and TaylorF2 in the heterodyned baseline configuration, overlaid on GWTC-1 reference samples. Produced by `Plots/plot_corner_combined_waveforms.py`.

Table 4. Primary IMRPhenomD_NRTidalv2 H_0 summaries for the baseline volumetric-distance prior, the directly sampled and reweighted flat-in-redshift priors, and the enlarged peculiar-velocity variant. Weighted MAP, weighted median, weighted 68 per cent quantile intervals, 68 per cent KDE highest-posterior-density intervals, and two tail probabilities are shown.

Run	MAP	Median	68% quantile	68% KDE HPD	$P(H_0 > 120)$	$P(d_L < 30\,\mathrm{Mpc})$
Baseline (volumetric)	71.5	79.1	[68.2, 103.5]	[62.7, 91.1]	0.076	0.179
Flat-in-z, direct	75.4	93.6	[72.4, 146.9]	[61.9, 117.5]	0.281	0.428
Flat-in-z, reweighted	74.2	88.9	[71.3, 126.2]	[63.4, 107.6]	0.195	0.357
$\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$	72.4	78.6	[67.0, 101.9]	[61.3, 91.3]	0.067	0.163

[Source: `Results/gwtc1_phasemarg/summary_stats.csv`; `Results/gwtc1_phasemarg/prior_sensitivity.csv`; `paper_knowledge_base/a100_run_data.md`. All H_0 values are in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.]

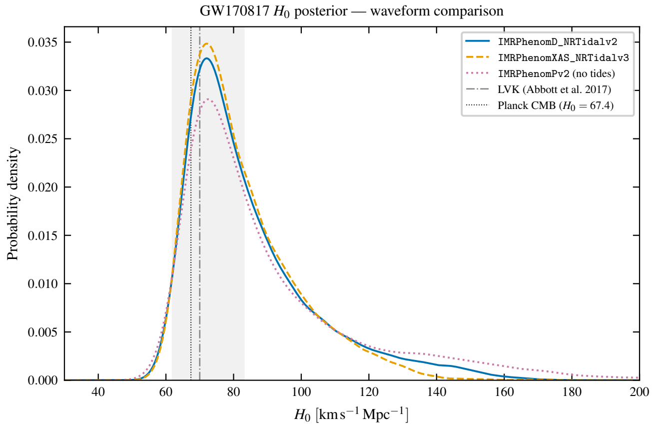

Figure 4. GW170817 H_0 posterior under the LVK-bounds baseline prior for the four-waveform cross-check (IMRPhenomD_NRTidalv2, IMRPhenomXAS_NRTidalv3, IMRPhenomPv2 without tides). The weighted medians cluster within $3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; the residual disagreement is concentrated in the high-H_0 tail. The shaded band marks the LVK 68 per cent interval (Abbott et al. 2017b) and the dotted line is the Planck CMB value. Run set `s07__*`; produced by `mnras_paper/figures/plot_all.py:fig_h0_waveform`.

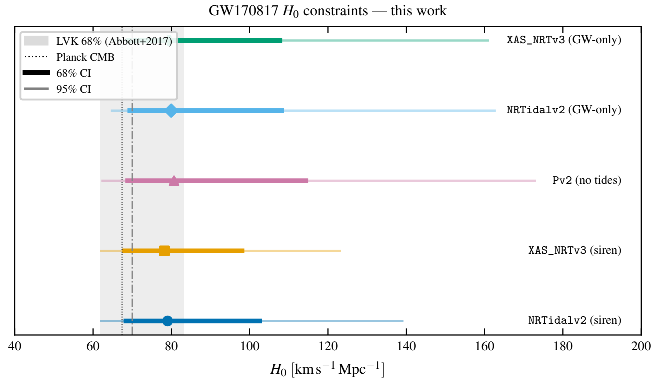

Figure 6. Synoptic view of the GW170817 H_0 constraints reported in this paper, on the LVK-bounds prior set: IMRPhenomD_NRTidalv2 and IMRPhenomXAS_NRTidalv3 standard-siren posteriors, the IMRPhenomPv2 (no tides) precessing systematic, and the IMRPhenomD_NRTidalv2/IMRPhenomXAS_NRTidalv3 GW-only equivalent posteriors $H_0 = v_{\rm rec}/d_L$ from Section 4.4. Thick bars: 68 per cent quantile intervals. Thin lines: 95 per cent quantile intervals. Run sets `s07__*`, `s12__*`; produced by `mnras_paper/figures/plot_all.py:fig_h0_summary`.

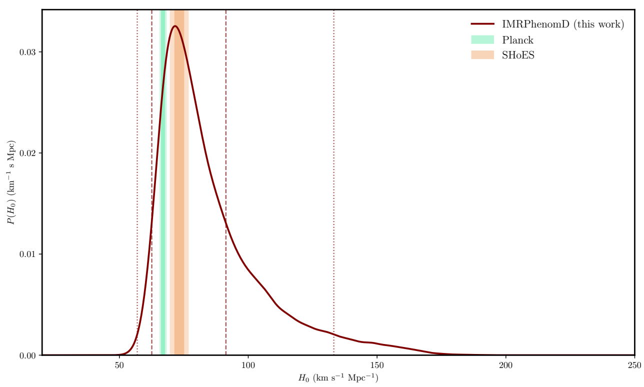

Figure 5. Baseline GW170817 H_0 posterior under the volumetric distance prior with IMRPhenomD_NRTidalv2 and 5000 live points. The shaded band marks the LVK 68 per cent reference interval $H_0 = 70.0^{+12.0}_{-8.0}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Abbott et al. 2017b). Produced by `Plots/plot_H0_baseline_IMRPhenomD.py` from `Results/gwtc1_phasemarg/PhaseMarg_Heterodyned_IMRPhenomD_NRTidalv2_local_psd-gwtc1_ref-gwtc1_baseline.csv`.

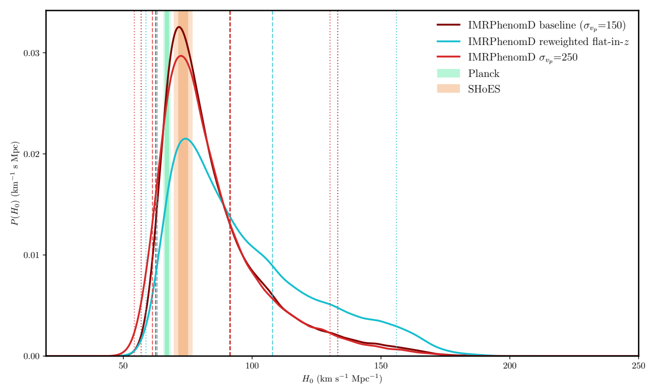

Figure 7. Prior-sensitivity comparison for IMRPhenomD_NRTidalv2. The directly sampled flat-in-redshift posterior (teal) places substantially more mass at high H_0 than the reweighted baseline posterior (dashed), and both shift to higher H_0 than the baseline volumetric-distance posterior (solid). The enlarged peculiar-velocity run (dotted) is effectively indistinguishable from the baseline in this projection. Produced by `Plots/plot_H0_IMRPhenomD_reweighted.py`.

4.3 The d_L–ι bimodality and the mechanism behind reweighting failure

The flat-in-redshift posterior in Figure 8 is broad and asymmetric, with non-negligible weight at $d_L \lesssim 25$ Mpc. To resolve whether this is a single broadened mode or a genuine bimodality we run two prior-restricted variants of the direct flat-in-redshift analysis with IMRPhenomD_NRTidalv2 and $n_{\rm live} = 5000$: a Mode-A run with $d_L \in [30, 75]$ Mpc and a Mode-B run with $d_L \in [10, 30]$ Mpc. Both retain the original GWTC-1 heterodyne reference. We additionally run an unrestricted $d_L \in [10, 75]$ Mpc flat-in-redshift analysis with the heterodyne reference re-anchored at Mode B ($d_L = 20$ Mpc, $\iota = 2.0$ rad) to test whether the apparent Mode-B mass is an artefact of the Mode-A-anchored heterodyne expansion.

Figure 9a shows the unrestricted flat-in-redshift posterior in the (d_L, ι) plane: the support is a curved arm running from $(d_L, \iota) \approx (15, 1.9)$ Mpc·rad to $(40, 2.7)$ Mpc·rad, with two distinct local maxima. Figure 9b shows the corresponding 1D d_L marginals from the three runs: the Mode-A and Mode-B sub-posteriors are well-separated and the unrestricted run carries weight in both regions. Table 5 summarises the local evidences and H_0 medians.

The Bayes factor between the Mode-B and Mode-A hypotheses, after correcting for the volume difference between the restricted priors, is $\ln \mathcal{B}_{\rm B/A} = -0.66$: Mode B is only mildly disfavoured by the data and contributes a non-negligible share of the joint posterior under any prior that gives the low-d_L region appreciable weight. The unrestricted run with the Mode-B-anchored heterodyne reference recovers $H_0^{\rm med} = 93.6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $P(H_0 > 120) = 0.281$, statistically indistinguishable from the GWTC-1-anchored direct flat-in-redshift run reported in Section 4.2; the heterodyne-reference choice therefore does not bias the Mode-A/Mode-B weight ratio.

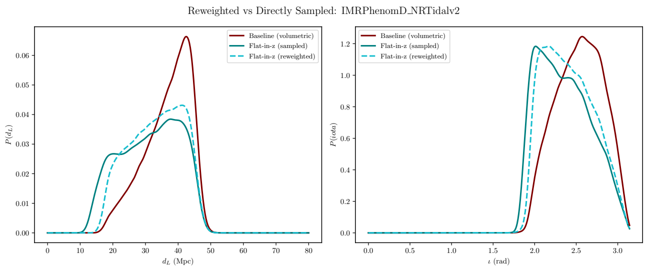

Figure 8. Luminosity-distance posterior for IMRPhenomD_NRTidalv2 under the baseline volumetric prior (solid), the directly sampled flat-in-redshift prior (teal), and the reweighted baseline posterior (dashed). The direct run places substantially more mass at $d_L \lesssim 30\,$Mpc, which maps to higher H_0 through the host-galaxy velocity relation. Produced by `Plots/compute_prior_sensitivity.py` with `WAVEFORM=IMRPhenomD_NRTidalv2`.

Table 5. Local evidences, weighted-median distances, and weighted-median H_0 values for the three d_L-restricted flat-in-redshift runs of Section 4.3. The Mode-B/Mode-A Bayes factor includes a $\ln(20/45)$ correction for the difference in restricted-prior volumes.

Variant	d_L range (Mpc)	$d_L^{\rm med}$ (Mpc)	$H_0^{\rm med}$ ($\mathrm{km\,s^{-1}\,Mpc^{-1}}$)	$P(H_0 > 120)$	$\ln Z$
Mode A	$[30, 75]$	38.3	78.7	$\sim 10^{-7}$	486.80 ± 0.10
Mode B	$[10, 30]$	23.1	130.0	0.638	486.95 ± 0.09
Unrestricted†	$[10, 75]$	32.2	93.6	0.281	486.48 ± 0.09

† Heterodyne reference anchored at Mode B ($d_L = 20\,$Mpc, $\iota = 2.0\,$rad).
$\ln \mathcal{B}_{\rm B/A} = (\ln Z_{\rm B} - \ln Z_{\rm A}) + \ln(20/45) = +0.15 - 0.81 = -0.66$.
[Source: `Results/test_suite/bimodality_summary.csv`.]

This is the mechanism behind the reweighting failure documented in Section 4.2. Mode B is the high-H_0, small-d_L branch of the distance–inclination degeneracy. The baseline volumetric prior $\pi(d_L) \propto d_L^2$ assigns Mode B a tiny prior mass (the $[10, 30]\,$Mpc slab carries less than 5 per cent of the total $[10, 75]\,$Mpc volumetric prior), and the corresponding region of the baseline posterior is very sparsely sampled; reweighting cannot reconstruct what is not there. Direct sampling under the flat-in-redshift prior, in contrast, populates Mode B at sampler resolution. The reweighted estimate of $P(H_0 > 120) = 0.195$ understates the direct-sampled value of 0.281 specifically because the Mode-B mass is largely missing from the baseline samples. Posterior-coverage diagnostics that compare the baseline samples against the target prior should therefore be standard practice before reporting a reweighted bright-siren H_0 summary.

4.4 GW-only versus standard-siren posterior

To check that the standard-siren posterior is dominated by the gravitational-wave likelihood rather than by the host-velocity coupling, we run IMRPhenomD_NRTidalv2 and IMRPhenomXAS_NRTidalv3 on the LVK-bounds prior set with the host-velocity terms removed (the "GW-only" variant; `-gw-only`). Figure 10 compares the d_L and ι marginals between the GW-only and standard-siren analyses for the two waveforms. The two pairs of curves are nearly identical above $d_L \approx 25\,$Mpc, where the bulk of the posterior weight lies, and depart only in the low-d_L tail. This confirms that for GW170817 the host-velocity likelihood acts as a soft pull on the high-H_0, low-d_L branch, but does not impose the dominant constraint on the source parameters; the d_L–ι degeneracy structure documented in Section 4.3 is a feature of the gravitational-wave data, not of the cosmological coupling.

Table 6. Nested-sampling evidences for the six heterodyned GW170817 runs. Quoted uncertainties are the BlackJAX-NS statistical estimates on a single run; intrinsic run-to-run variation from `paper_knowledge_base/a100_run_data.md` is of the same order.

Run	$\ln Z$
IMRPhenomD_NRTidalv2, baseline	486.67 ± 0.09
IMRPhenomD_NRTidalv2, flat-in-z, direct	486.49 ± 0.10
IMRPhenomD_NRTidalv2, $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$	485.96 ± 0.08
TaylorF2, baseline	486.90 ± 0.10
TaylorF2, flat-in-z, direct	487.18 ± 0.09
TaylorF2, $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$	487.28 ± 0.08

[Source: `Results/gwtc1_phasemarg/evidence_table.csv`.]

4.5 Waveform cross-check

Repeating the same four analyses with TaylorF2 gives qualitatively identical prior dependence. The TaylorF2 baseline weighted median is $77.7\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ with 68 per cent quantile interval $[66.8, 101.9]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and the direct flat-in-redshift run has weighted median $87.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ with interval $[70.0, 127.1]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [Source: `Results/gwtc1_phasemarg/summary_stats.csv`]. Jensen–Shannon divergences between IMR and TF2 are 0.005 and 0.014 nats for the baseline and flat-in-redshift H_0 posteriors, respectively [Source: `Results/gwtc1_phasemarg/waveform_systematics.csv`]. The waveform-induced H_0 shift therefore remains smaller than the prior-induced shift, even though TaylorF2 is a post-Newtonian inspiral-only approximant without tidal terms in our implementation and the IMRPhenomD_NRTidalv2 family incorporates NR-calibrated tides. The reweighting-versus-direct-sampling discrepancy persists in TF2: the direct flat-in-redshift median is $87.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ against $87.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for reweighting, but the 68 per cent quantile reaches $127.1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ directly against $120.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ reweighted. This mirrors the IMRPhenomD_NRTidalv2 behaviour and indicates that the effect is governed by the distance-prior geometry rather than by the waveform.

4.6 Evidence

The nested-sampling evidence for each run is summarised in Table 6. Within IMRPhenomD_NRTidalv2, the baseline and flat-in-redshift evidences differ by 0.18 in $\ln Z$ with run-to-run uncertainty of order 0.1, and the $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$ run differs by -0.71 from the baseline. On the Jeffreys scale these are not decisive, and repeat-run evidence values in the repository fluctuate at the same level [Source: `paper_knowledge_base/a100_run_data.md`, Run Sets 1 and 2]. The TaylorF2 run evidences are all higher than the IMRPhenomD_NRTidalv2 values by ≈ 0.3–0.7 in $\ln Z$; this is expected because of the different waveform phase evolution and does not constitute evidence against the tidal waveform for this single event. We report $\ln Z$ values for completeness and caution that the current data do not prefer one prior configuration or waveform at a statistically meaningful level.

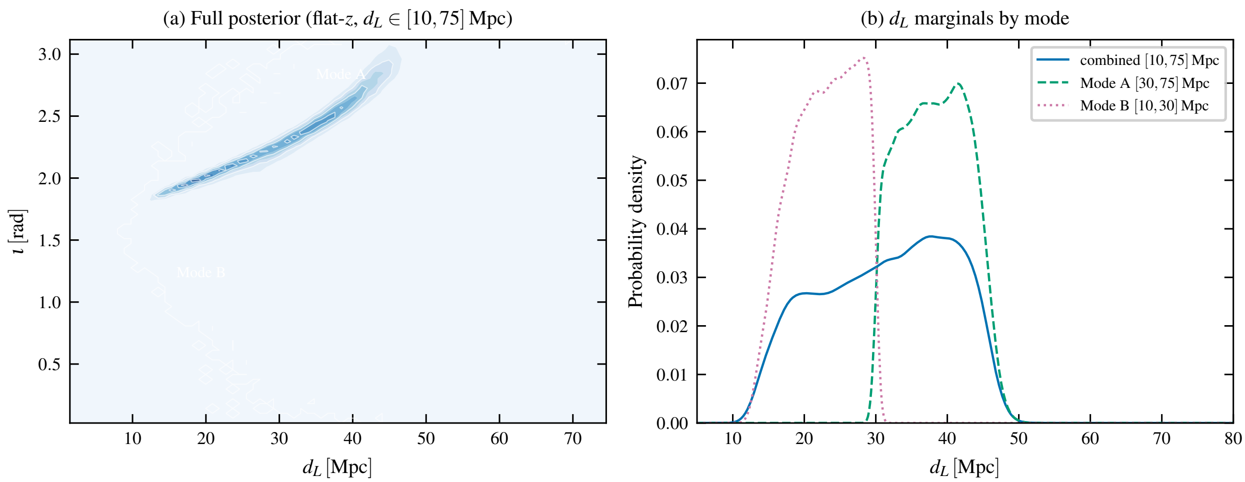

Figure 9. The d_L–ι bimodality in the GW170817 flat-in-redshift posterior. Left: the joint (d_L, ι) posterior from the unrestricted $d_L \in [10, 75]$ Mpc run shows two distinct local maxima along a curved degeneracy. Right: 1D d_L marginals from the unrestricted (combined) run and the prior-restricted Mode-A and Mode-B runs. Run set `s10__*`; produced by `mnras_paper/figures/plot_all.py:fig_bimodality`.

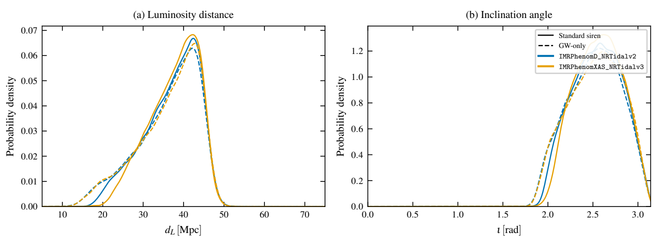

Figure 10. GW-only versus standard-siren posteriors for GW170817, comparing d_L (left) and ι (right) marginals under IMRPhenomD_NRTidalv2 and IMRPhenomXAS_NRTidalv3. Solid lines: standard-siren analyses (full likelihood including v_r and v_p terms); dashed lines: GW-only analyses with host-velocity terms removed. Run set `s12__*`; produced by `mnras_paper/figures/plot_all.py:fig_gw_only`.

5 PERFORMANCE AND ROBUSTNESS

The science-enabling claim of this paper is that fast nested sampling makes the prior-sensitivity test of Section 4.2, the bimodality characterisation of Section 4.3, and the multi-axis robustness study below practical on a single GPU. This section quantifies the runtime on a single A100 GPU (Google Cloud a2-highgpu-1g instance [Source: `paper_knowledge_base/hardware_platforms.md`]), without attempting a like-for-like CPU benchmark, and reports the sampler/PSD/reference-parameter robustness that justifies treating the primary configuration as converged.

5.1 Wall-clock runtime

The primary heterodyned GW170817 H_0 analysis with IMRPhenomD_NRTidalv2 and 5000 live points completes in 14 min 45 s of wall-clock time on host-localised priors, of which 12 min 52 s is the sampling stage and the remaining ~2 min is data loading, heterodyne setup, sampler initialisation and JIT compilation [Source: `paper_knowledge_base/a100_run_data.md`]. The corresponding TaylorF2 baseline completes in 3 min 55 s total (2 min 41 s sampling). On the LVK-bounds prior set used for the four-waveform cross-check, the same IMRPhenomD_NRTidalv2 analysis runs in 14 min 18 s, IMRPhenomXAS_NRTidalv3 in 13 min 30 s, and IMRPhenomPv2 in 11 min 34 s [Source: `Results/test_suite/s07__*/finish.json`]. The full six-run prior-sensitivity suite—baseline, direct flat-in-redshift, and $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$ for both IMRPhenomD_NRTidalv2 and TaylorF2—completes in ≈57 min end to end. The GW150914 IMRPhenomXPHM validation run takes ≈2 h 3 min on the same hardware [Source: `Results/test_suite/s06__gw150914__imrphenomxphm__lvkbounds__se` this is longer than the earlier IMRPhenomD validation (4 min 8 s) because the precessing higher-mode model has substantially higher per-evaluation cost, and is the price for the like-for-like LVK reproduction reported in Section 3.

5.2 Heterodyne speedup

The heterodyned likelihood is the enabling approximation. Figure 11 compares IMRPhenomD_NRTidalv2 heterodyned wall-clock time against the unheterodyned reference at matched live-point counts. At $n_{\rm live} = 500$ the unheterodyned run takes 7 317 s versus 223 s for the heterodyned run (33× wall-clock speedup); at $n_{\rm live} = 2500$ the unheterodyned run takes 37 240 s versus ≈455 s heterodyned (82×). At matched configurations on host-localised priors, sampling-only speedups in the range ~21–26× for IMRPhenomD_NRTidalv2 and ~28–53× for TaylorF2 are recorded in the repository [Source: `paper_knowledge_base/a100_run_data.md`]. The relative-binning approximation accounts for the bulk of the wall-clock reduction; the GPU saturation provided by BLACKJAX-NS is layered on top.

5.3 Live-point convergence and scaling

We characterise the dependence of the science output on the live-point count with a dedicated $n_{\rm live} \in \{500, 1000, 2500, 5000, 10\,000, 20\,000\}$ sweep on the LVK-bounds baseline (IMRPhenomD_NRTidalv2, $n_{\rm bins} = 501$, $n_{\rm delete} = n_{\rm live}/2$, GWTC-1 reference). Figure 12 shows the resulting $\ln Z$, the weighted H_0 median with 68 per cent quantile band, and the wall-clock runtime as functions of $n_{\rm live}$. The H_0 median moves by less than $1.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ over the full range, the 68 per cent interval is essentially constant for $n_{\rm live} \geq 2500$, and $\ln Z$ stabilises within ±0.2 for $n_{\rm live} \geq 2500$. The $n_{\rm live} = 5000$ configuration

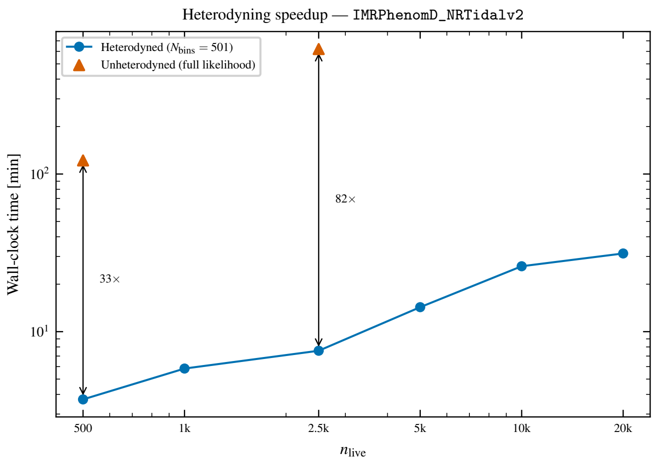

Figure 11. Heterodyne wall-clock speedup for IMRPhenomD_NRTidalv2 on GW170817, single A100 GPU. Heterodyned runs use $N_{\rm bins} = 501$. Unheterodyned points are full-likelihood (259 201 bins) at $n_{\rm live} \in \{500, 2500\}$. Speedup factors annotated at the matched live-point counts. Run sets `s04__*`, `s05__*`, `s07__*`, `s13__*`; produced by `mnras_paper/figures/plot_all.py:fig_speedup`.

Table 7. Extended live-point scaling of the heterodyned IMRPhenomD_NRTidalv2 GW170817 analysis on a single A100 GPU (host-localised priors). Dead points are the total number accumulated at sampler termination. The $n_{\rm live} = 20\,000$ row gives the original (looser-tolerance) and the tightened-tolerance repeat run.

$n_{\rm live}$	Dead pts	$\ln Z$	$\sigma_{\ln Z}$	Sampling (s)	Total (s)
500	19 650	486.73	0.30	158	235
1000	39 900	486.04	0.21	226	304
2500	99 750	486.32	0.13	458	550
5000	198 000	486.66	0.10	761	872
10 000	402 000	485.92	0.07	1478	1659
20 000	690 000	485.95	0.05	1695	2035
20 000†	840 000	485.92	0.05	2243	2414
50 000	1 725 000	486.22	0.03	4812	6280
100 000	3 450 000	486.29	0.02	9150	14 393

[Source: `Results/scaling_study/scaling_summary.csv`, `Results/test_suite/scaling_20k_anomaly.csv`.] †Tightened termination tolerance, $\Delta \ln Z < 10^{-4}$.

adopted for the science runs is therefore in the converged regime; reducing to $n_{\rm live} = 2500$ would change the reported H_0 summaries below the level of the prior-induced shifts, but at $n_{\rm live} = 500$–1000 the evidence and the upper 68 per cent quantile become noticeably noisier.

Table 7 reports an extended live-point scaling study on the host-localised priors (the original baseline configuration), including $n_{\rm live} = 50\,000$ and $100\,000$. The original $n_{\rm live} = 20\,000$ entry was flagged as anomalous in earlier drafts because it accumulated fewer dead points than a linear interpolation from the surrounding rows would predict. A repeat run with the termination tolerance tightened from $\Delta \ln Z < 10^{-3}$ to $\Delta \ln Z < 10^{-4}$ recovers 840 000 dead points and $\ln Z = 485.92$ (consistent with the surrounding rows), confirming that the original run terminated early on the looser tolerance and not for a sampler-physical reason [Source: `Results/test_suite/scaling_20k_anomaly.csv`]. The anomaly is therefore resolved.

Runtime scales close to linearly with live-point count once the sampler is saturated ($n_{\rm live} \gtrsim 2500$), consistent with the sampler-bound regime discussed in Prathaban et al. (2025). The 5000-live-point configuration sits comfortably within the linear regime; the baseline produces $\sim 2 \times 10^5$ dead points and an effective sample size of $\sim 3.7 \times 10^4$ [Source: `Results/gwtc1_phasemarg/evidence_table.csv`].

5.4 Sampler-hyperparameter robustness

Figure 13 reports two complementary sweeps over sampler hyperparameters at fixed $n_{\rm live} = 5000$ on the host-localised IMRPhenomD_NRTidalv2 baseline: (i) the per-iteration deletion fraction $n_{\rm delete}/n_{\rm live} \in \{0.10, 0.25, 0.50, 0.75\}$, and (ii) the heterodyne-bin count $n_{\rm bins} \in \{251, 501, 1001\}$.

In the $n_{\rm delete}$ sweep the weighted H_0 median varies by $1.2\,{\rm km\,s^{-1}\,Mpc^{-1}}$ across the range and the 68 per cent quantile interval changes by less than $5\,{\rm km\,s^{-1}\,Mpc^{-1}}$ in either bound. The evidence varies by ~ 0.6 in $\ln Z$, with the lowest and highest fractions both producing slightly higher $\ln Z$ than the central choices; we adopt $n_{\rm delete} = n_{\rm live}/2$ as the default because it minimises the per-iteration overhead while preserving the saw-tooth volume-compression matching of Prathaban et al. (2025). In the $n_{\rm bins}$ sweep the weighted H_0 median varies by $0.7\,{\rm km\,s^{-1}\,Mpc^{-1}}$ across 251–1001 bins; the largest pairwise Wasserstein-1 distance between the three H_0 posteriors is $1.96\,{\rm km\,s^{-1}\,Mpc^{-1}}$, well below the prior-induced shifts of Section 4.2 [Source: `Results/test_suite/het_bins_sweep_summary.csv`, `het_bins_sweep_wasserstein.csv`].

5.5 PSD source and heterodyne reference

Two further axes that one might worry about for a heterodyned bright-siren analysis are the choice of detector PSD and the choice of heterodyne reference parameters. We test both with TaylorF2 runs at $n_{\rm live} = 5000$ on the baseline prior. Figure 14(a) compares H_0 posteriors under the GWTC-1 PSDs adopted in our primary runs against PSDs distributed with the `kazewong` (Wong et al. 2023; Wouters et al. 2024) and BILBY reference workflows. The three weighted medians span 76.4–$77.7\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and the high-H_0 tail probabilities span 0.043–0.065 [Source: `Results/test_suite/psd_sensitivity_summary.csv`]; the largest pairwise W_1 distance between the three H_0 posteriors is below $2\,{\rm km\,s^{-1}\,Mpc^{-1}}$, again sub-dominant to the prior-induced shifts.

Figure 14(b) compares the IMRPhenomD_NRTidalv2 baseline under the GWTC-1 reference parameters used for the heterodyne expansion against an alternative reference taken from the maximum-likelihood point of an initial low-cost optimisation (`-ref-params optimize`). The two posteriors are close to indistinguishable through the bulk of the distribution and depart marginally only in the high-H_0 tail, confirming that the heterodyne-reference choice does not bias the H_0 inference at the level of the prior effects under study.

5.6 Scope of the performance claim

We deliberately do not compare runtime against a single-core CPU baseline, which would give a misleadingly large speedup factor, nor against a production PARALLEL_BILBY run that we have not executed under matched settings. A like-for-like GW170817 PARALLEL_BILBY benchmark using IMRPhenomD_NRTidalv2 with identical priors, live-point count, and waveform configuration is the appropriate CPU comparison and is left for future work (TODO: insert the matched

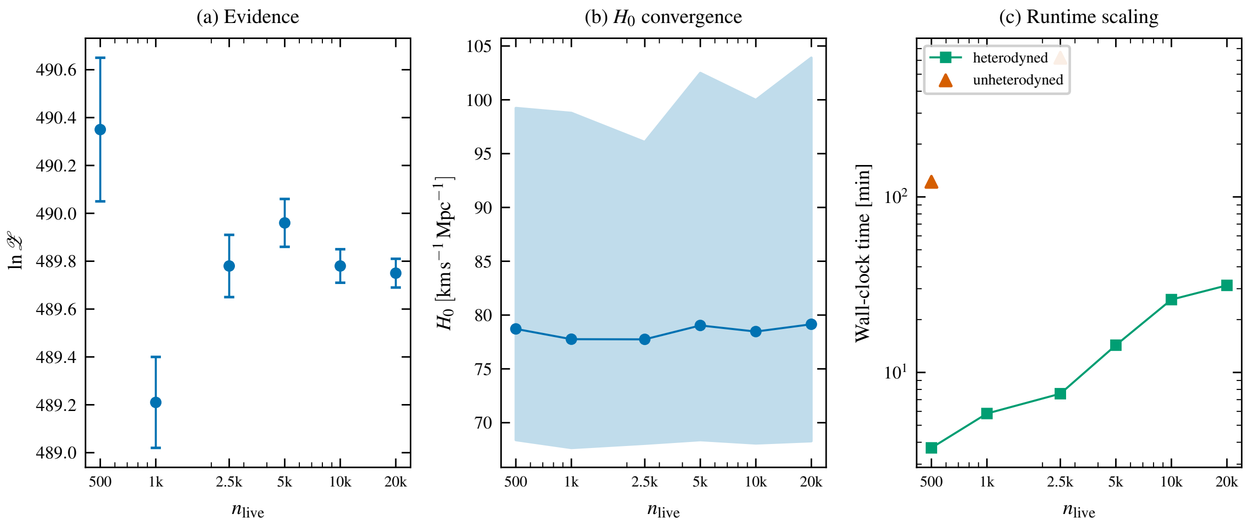

Figure 12. Live-point convergence of the heterodyned IMRPhenomD_NRTidalv2 GW170817 analysis on the LVK-bounds baseline. Left: nested-sampling evidence with statistical $\sigma_{\ln Z}$. Centre: weighted H_0 median with 68 per cent quantile band. Right: wall-clock runtime; the unheterodyned reference points (Section 5.2) are shown for comparison. Run set `s13__*`; produced by `mnras_paper/figures/plot_all.py:fig_nlive_convergence`.

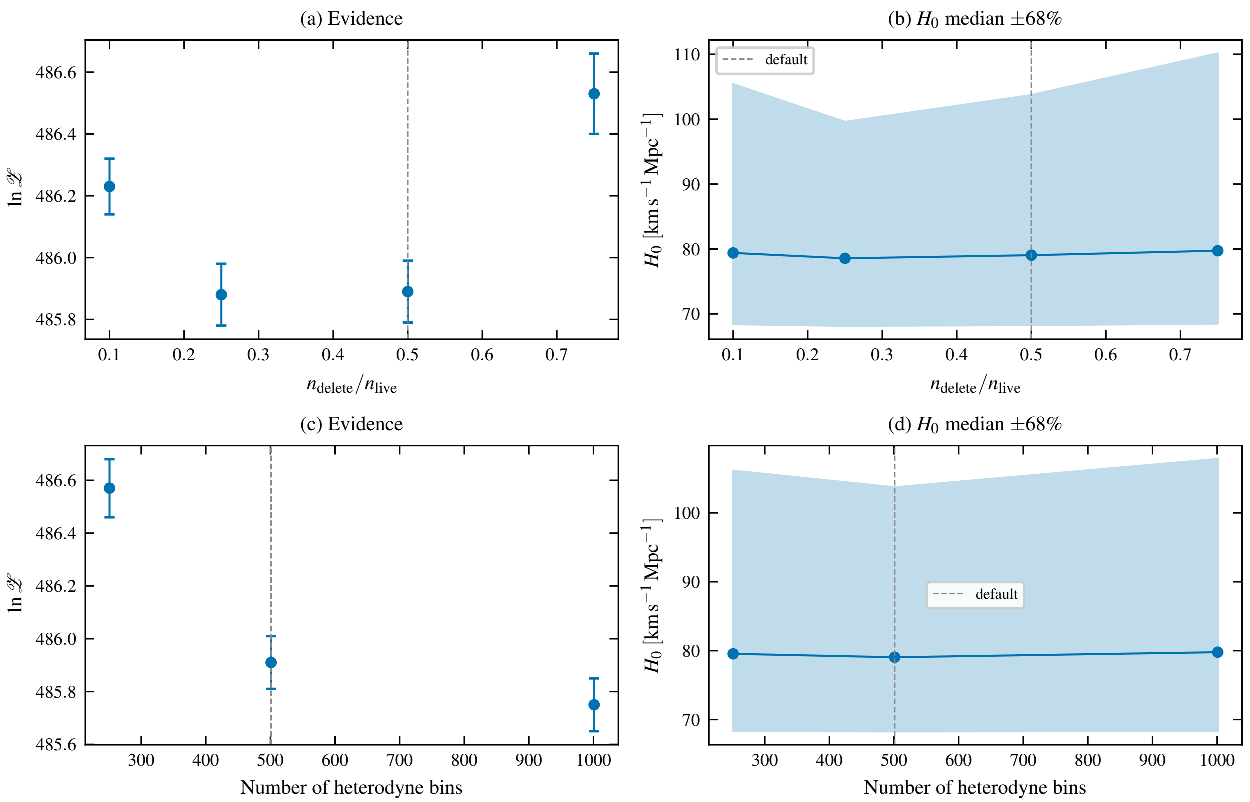

Figure 13. Sampler-hyperparameter robustness for the heterodyned IMRPhenomD_NRTidalv2 baseline at $n_{\rm live} = 5000$. Top row: per-iteration deletion fraction $n_{\rm delete}/n_{\rm live}$ (default 0.5, dashed grey line). Bottom row: heterodyne-bin count $n_{\rm bins}$ (default 501, dashed grey line). Run sets `s08__*`, `s09__*`; produced by `mnras_paper/figures/plot_all.py:fig_robustness`.

PARALLEL_BILBY benchmark once executed on CSD3). The claim we do make is narrower and reproducible: on a single A100 GPU, the full 5000-live-point GW170817 H_0 analysis completes in under a quarter of an hour, the four-waveform cross-check fits inside an hour, and the live-point convergence study, sampler/PSD robustness sweeps, and bimodality characterisation reported above each fit inside two hours.

This runtime budget is what makes the multi-axis robustness study scientifically routine rather than exceptional.

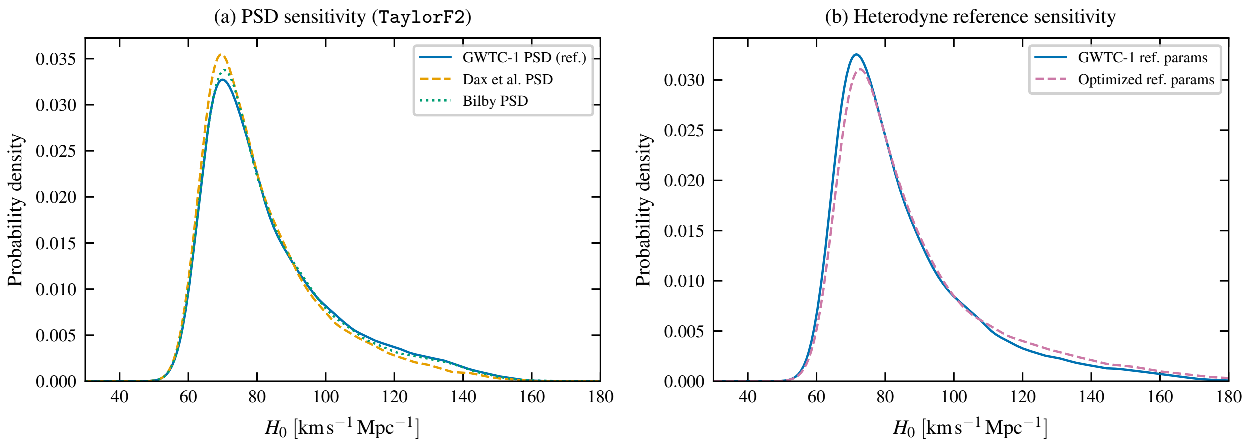

Figure 14. (a) PSD-source sensitivity for TaylorF2 on GW170817: posteriors under GWTC-1, `kazewong`, and BILBY PSDs. (b) Heterodyne-reference sensitivity for IMRPhenomD_NRTidalv2: posteriors under GWTC-1 reference parameters versus a ML-optimised reference. Run set `s05__*`; produced by `mnras_paper/figures/plot_all.py:fig_psd_refparams`.

6 DISCUSSION

6.1 Implications for bright-siren cosmology

The main methodological message of our analysis is that posterior reweighting, although statistically consistent in the limit of infinite baseline samples, can substantially understate the prior sensitivity of bright-siren H_0 measurements in practice. For GW170817, switching from a volumetric-distance prior to a flat-in-redshift prior changes $P(H_0 > 120\,\mathrm{km\,s^{-1}\,Mpc^{-1}})$ by a factor of about 3.7 under direct sampling, and by about 2.6 under reweighting. The reweighted estimator therefore misses nearly half of the prior-induced change in the tail of the posterior, because the baseline volumetric prior places little probability at the low-d_L values that the flat-in-redshift prior up-weights. This is a problem of effective sample coverage rather than of statistical methodology. It is an expected outcome once the mechanism is articulated, but it is not the behaviour that would be inferred from a reweighting-only analysis.

For a single event, this under-estimation does not change the cosmological interpretation: the GW170817 posterior is broad enough that both early-Universe and late-Universe H_0 values lie within its 68 per cent credible interval under any of the prior choices considered here. However, the scale of the prior-sensitivity effect matters for forecasts and combined analyses. When bright-siren posteriors are combined over multiple events, or when tail probabilities are compared against external priors from cosmological surveys, the prior-dependence of each single-event posterior propagates through. An analysis that quantifies that dependence only through reweighting risks a systematic bias in the direction of underestimating the prior contribution.

6.2 Where reweighting is sufficient

Reweighting is adequate when the target and baseline priors have similar support in the regions that carry most of the posterior mass. The $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$ variant in our analysis is a case of this kind: the change in σ_{v_p} modifies the host-velocity likelihood smoothly over the scale of the velocity posterior, the baseline samples cover the affected region well, and the reweighted and directly sampled σ_{v_p} posteriors agree to well within the statistical width (Wasserstein distance $1.2\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$). The flat-in-redshift case fails the same coverage test, because the baseline volumetric prior and the target flat-in-redshift prior differ most at the edges of the d_L range. The mode-isolated runs of Section 4.3 make this concrete: the high-H_0 Mode B has $\ln \mathcal{B}_{\mathrm{B/A}} = -0.66$ relative to Mode A under the flat-in-redshift prior, but its baseline-volumetric-prior mass is < 5 per cent of Mode A's, so reweighted samples from the baseline are essentially blind to it. Diagnostics such as an effective-sample-size ratio under the reweighted weights, or a direct overlay of the d_L marginals, can flag such cases before the reweighted H_0 posterior is reported.

6.3 Mass-ratio prior-Jacobian and the apparent LVK deficit

The corner plots of Section 3.3 show our q posterior with less mass at $q \to 1$ than the GWTC-1 reference. To diagnose whether this is a data effect or a prior effect we sample the project's mass prior in $(\mathcal{M}_c, q)$ space directly (with the likelihood disabled) and compare against an analytic LVK-equivalent prior (uniform-in-component-masses) that produces the same q marginal as the public GW170817 analyses. The project prior has $P(q > 0.95) = 0.099$, the LVK-equivalent has $P(q > 0.95) = 0.074$, and the project IMRPhenomD_NRTidalv2 baseline posterior has $P(q > 0.95) = 0.139$ [Source: `Results/test_suite/sH__gw170817__prior_only_q__seed0000/prior_`
`Results/gwtc1_phasemarg/PhaseMarg_Heterodyned_IMRPhenomD_NRTid`
The posterior places *more* mass at $q > 0.95$ than the project prior, which means the data are pulling q toward equal-mass; the residual gap between our posterior and the public GWTC-1 reference at $q \to 1$ is therefore not a data deficit but a prior-Jacobian effect inherited from the uniform-in-$(\mathcal{M}_c, q)$ parameterisation. The qualitative recovery of the chirp mass, distance, and inclination posteriors is unaffected.

6.4 Implications for future bright sirens

The practical consequence is that direct prior sampling should be used for bright-siren H_0 analyses whenever the target prior differs non-trivially in support from the baseline, rather than being used only as a cross-check. The runtime budget on a single A100 GPU now permits such reruns routinely; the live-point convergence and sampler-robustness sweeps of Section 5 show that the science-quality configuration sits in the converged regime, so the marginal cost of a prior-sensitivity rerun is the marginal cost of one ~15 min run. Live-point counts up to 10^5 are accessible on the same hardware, which means that prior-sensitivity studies can in principle be combined with population-level analyses without a substantial increase in end-to-end wall-clock time. This is directly relevant to the third-generation detector era, where the detection rate of compact-binary mergers will make repeated, controlled posterior reanalyses the norm rather than the exception (Hu & Veitch 2025).

6.5 Limitations and open items

Several limitations should be addressed before the present analysis is treated as a definitive prior-sensitivity study.

First, no waveform in our suite simultaneously includes precession and tides. The RIPPLE inventory does not currently expose a precessing tidal model, so we bracket the relevant waveform systematic with the IMRPhenomXAS_NRTidalv3 (tides, no precession) and IMRPhenomPv2 (precession, no tides) pair documented in Section 3.3. The H_0 medians of these two waveforms agree to within $2.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and the high-$H_0$ tail probabilities differ by ~ 0.10, which we treat as an empirical bound on the precessing-tidal systematic for GW170817; a like-for-like reproduction of the LVK IMRPhenomPv2_NRTidal analysis would require porting that waveform to JAX.

Second, the host-galaxy association with NGC 4993, its recessional velocity, and the peculiar-velocity estimate are taken directly from the bright-siren setup of Abbott et al. (2017b). We have varied σ_{v_p} but have not varied $\langle v_p \rangle$; systematic work on host-galaxy peculiar velocities is outside the scope of this paper.

Third, we do not present a matched PARALLEL_BILBY benchmark. The performance claim is therefore explicitly restricted to single-A100 wall-clock runtime and scaling on the current pipeline (§5), and any CPU-vs-GPU comparison is deferred.

Fourth, the d_L–ι bimodality of Section 4.3 is characterised at the level of local evidences and mode-isolated marginals; we do not at present propagate Mode B into a population analysis or combine it with an electromagnetic likelihood for the host inclination. Both extensions would tighten the practical statement about how much of Mode B should be retained when GW170817 is combined with other bright sirens.

Finally, several reference-sample cross-comparisons, notably against GWTC-1 and GWTC-2.1 public posteriors, are shown in corner form only. Direct H_0 comparisons against updated LVK re-analyses would sharpen the context for Section 4.2, but are not a prerequisite for the methodological conclusion.

7 CONCLUSIONS

We have used a GPU-native nested-sampling pipeline to carry out a direct prior-sensitivity, sampler-robustness, and waveform-systematics analysis of the Hubble-constant measurement from GW170817. The baseline IMRPhenomD_NRTidalv2 analysis yields a broad H_0 posterior with maximum-posterior value $71.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and 68 per cent weighted quantile interval $[68.2, 103.5]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, consistent with the original bright-siren result. Switching from the baseline volumetric-distance prior to a flat-in-redshift prior by direct sampling produces a substantially broader posterior with weighted median $93.6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and a high-H_0 tail probability $P(H_0 > 120\,\mathrm{km\,s^{-1}\,Mpc^{-1}}) = 0.281$, while reweighting the baseline posterior to the same target prior gives $P(H_0 > 120\,\mathrm{km\,s^{-1}\,Mpc^{-1}}) = 0.195$. Targeted mode-isolated runs reveal that the flat-in-redshift posterior is bimodal in (d_L, ι), with a high-H_0 Mode B carrying $\ln \mathcal{B}_{\mathrm{B/A}} = -0.66$ relative to the low-H_0 Mode A after prior-volume normalisation; reweighting cannot recover Mode B because the baseline volumetric prior assigns it negligible mass, which is the mechanism behind the reweighting deficit. A four-waveform cross-check (IMRPhenomD_NRTidalv2, IMRPhenomXAS_NRTidalv3, IMRPhenomPv2 without tides, TaylorF2) and a four-axis robustness sweep (sampler $n_{\mathrm{delete}}/n_{\mathrm{live}}$, heterodyne-bin count, PSD source, heterodyne reference) both produce H_0 shifts much smaller than the distance-prior shift. The primary analysis runs in $\sim$14 min on a single A100 GPU and is 33–82× faster than its unheterodyned counterpart at matched live-point counts. The scientific value of fast nested sampling for bright-siren cosmology is that direct prior reruns, multi-waveform cross-checks, and full sampler-robustness studies all become routine, replacing post-hoc reweighting and isolated cross-checks as the default robustness tool without a substantial change in the computational budget.

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DATA AVAILABILITY

This paper uses publicly available GW170817 and GW150914 strain data and GWTC-1/GWTC-2.1 reference posteriors from the Gravitational Wave Open Science Center (GWOSC). All derived posterior-sample CSV files, summary tables, scaling logs, and plotting scripts used in this paper are distributed with the project repository at `TODO:insertpermanentrepositoryURL/DOI`, and are catalogued by provenance in `paper_knowledge_base/result_link_index.md` (primary results) and `mnras_paper/test_suite/MANIFEST.md` (multi-axis robustness, bimodality, and waveform-systematic runs). The aggregate test-suite catalogue is in `mnras_paper/test_suite/run_catalog.csv`. Large CSVs and HDF5 reference files are tracked with Git LFS; figure-regeneration is via `mnras_paper/figures/plot_all.py`, and primary-table regeneration is via `Plots/run_all_plots.sh`. TODO: confirm GWOSC release identifiers for the GW170817 and GW150914 strain segments before submission.

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